



CHRISTOPHER ASH

Robotics and AI R&D Engineer

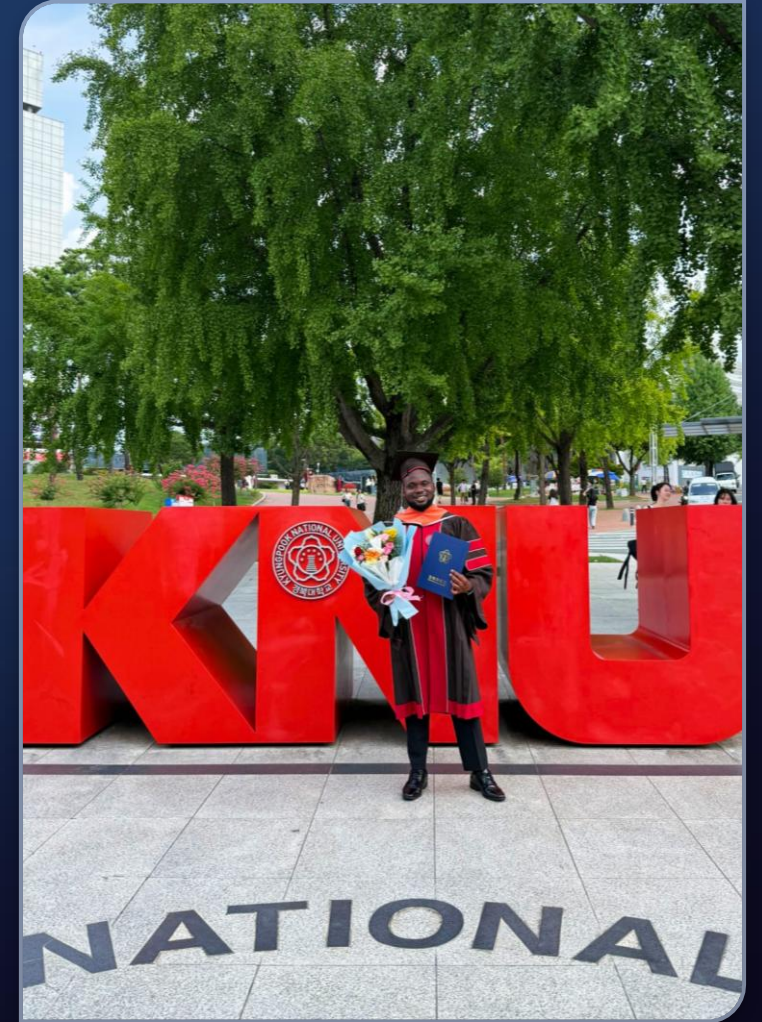
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SELF INTRODUCTION

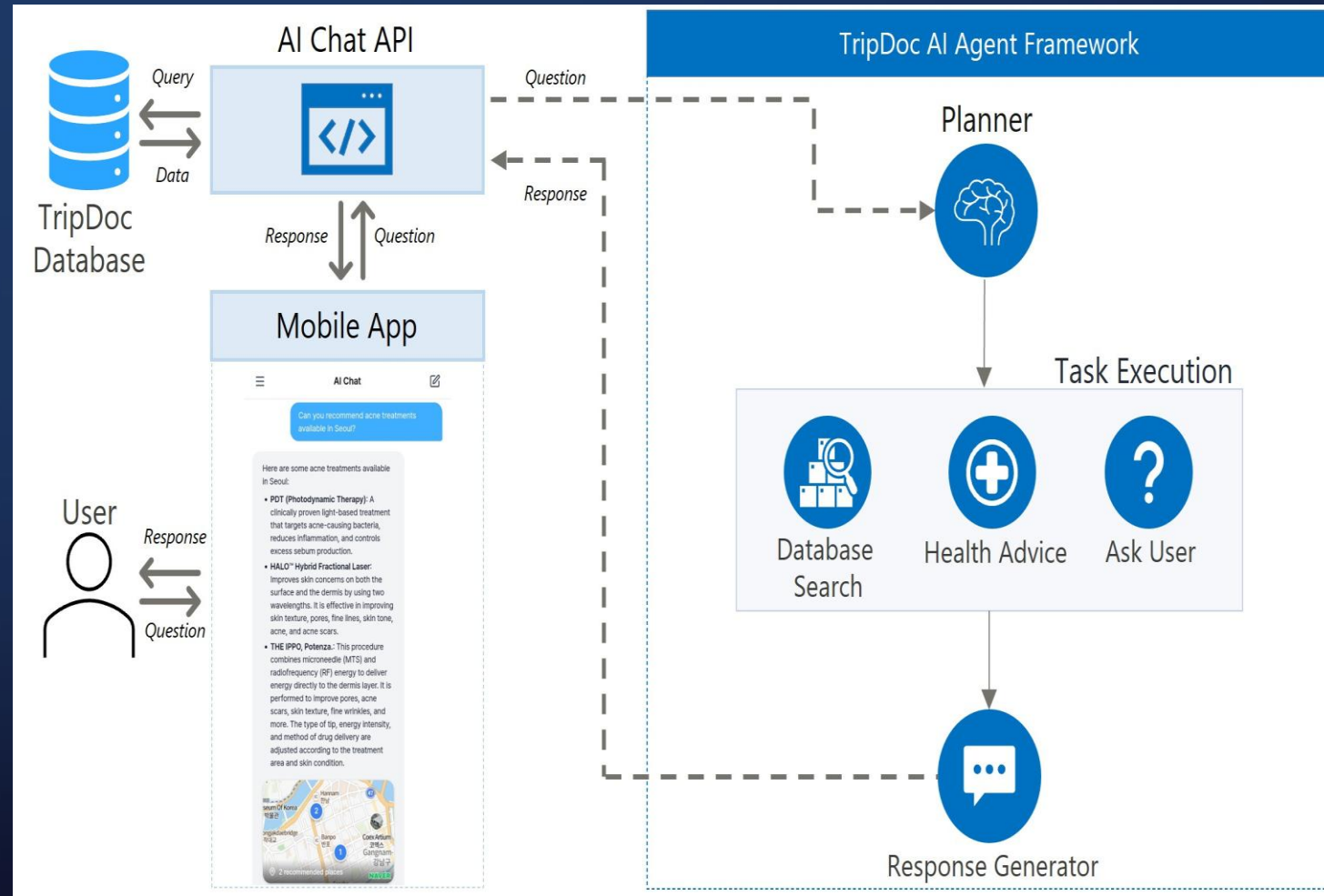
- **Name:** Christopher Ash
- **Nationality:** Trinidad and Tobago
- **Bachelor's Degree:** Electrical and Computer Engineering (The University of the West Indies, Trinidad)
- **Master's Degree:** Robotics and Smart Systems Engineering + AI Safety Convergence (Kyungpook National University)
- **Hobbies:** Chess, Trying new foods, Weightlifting



R&D PROJECT:

TRIPDOC AI CHAT ASSISTANT

- An agentic, multilingual LLM assistant for a medical-tourism platform allowing users to search hospitals, treatments, and products and move through booking entirely in natural language.
- Built on a planner-task architecture: an LLM classifies intent and generates an executable plan, run against real data sources including a SQL agent over PostgreSQL and external APIs, separating decision logic from data access.
- Multi-turn booking flow driven by a deterministic stage controller, with guardrails placed throughout the system for security, correctness, and safe medical-domain behaviour.

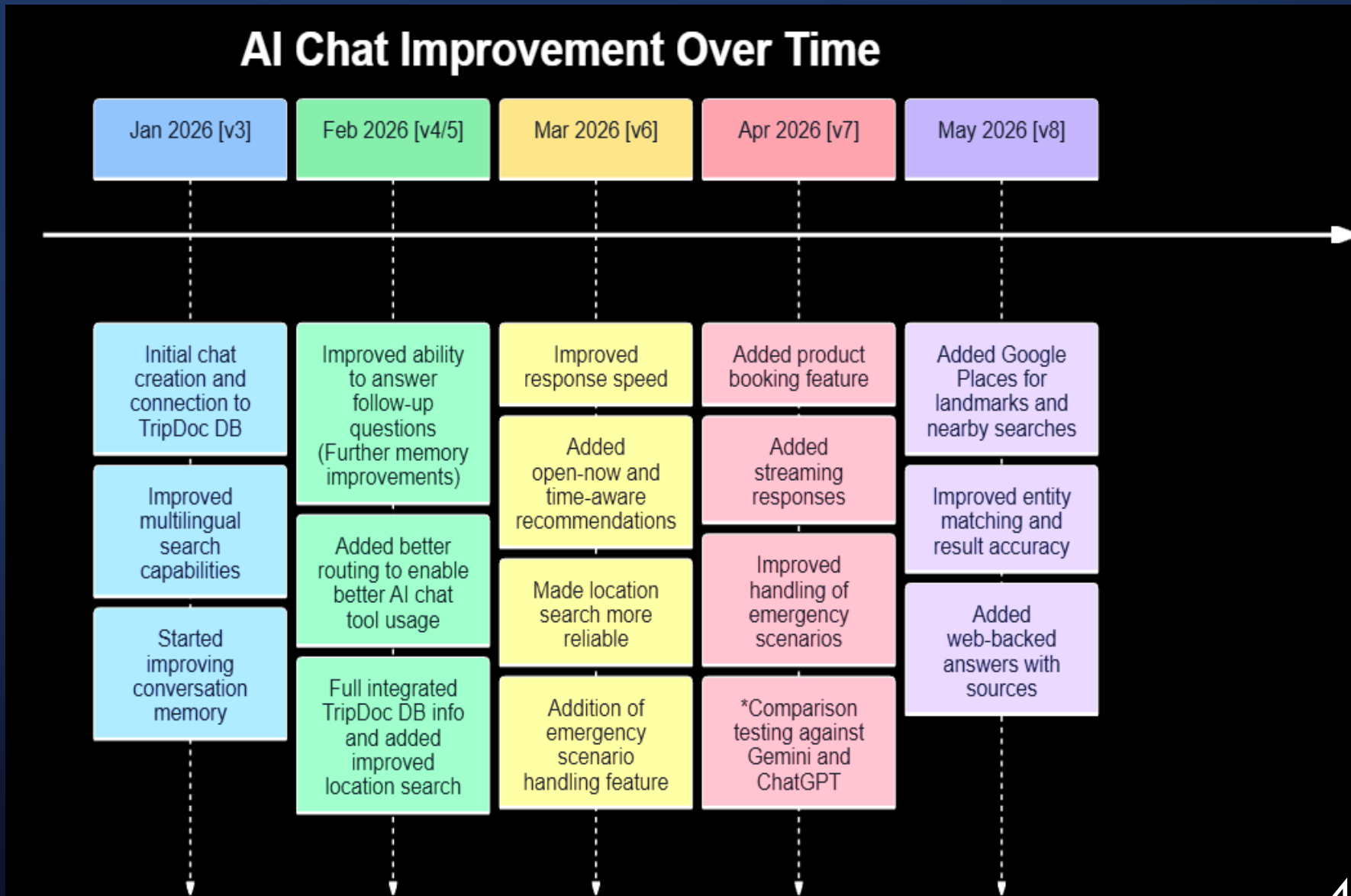


PROJECT TIMELINE:

TRIPDOC AI CHAT ASSISTANT

FEATURE PROGRESSION: JAN–MAY 2026.

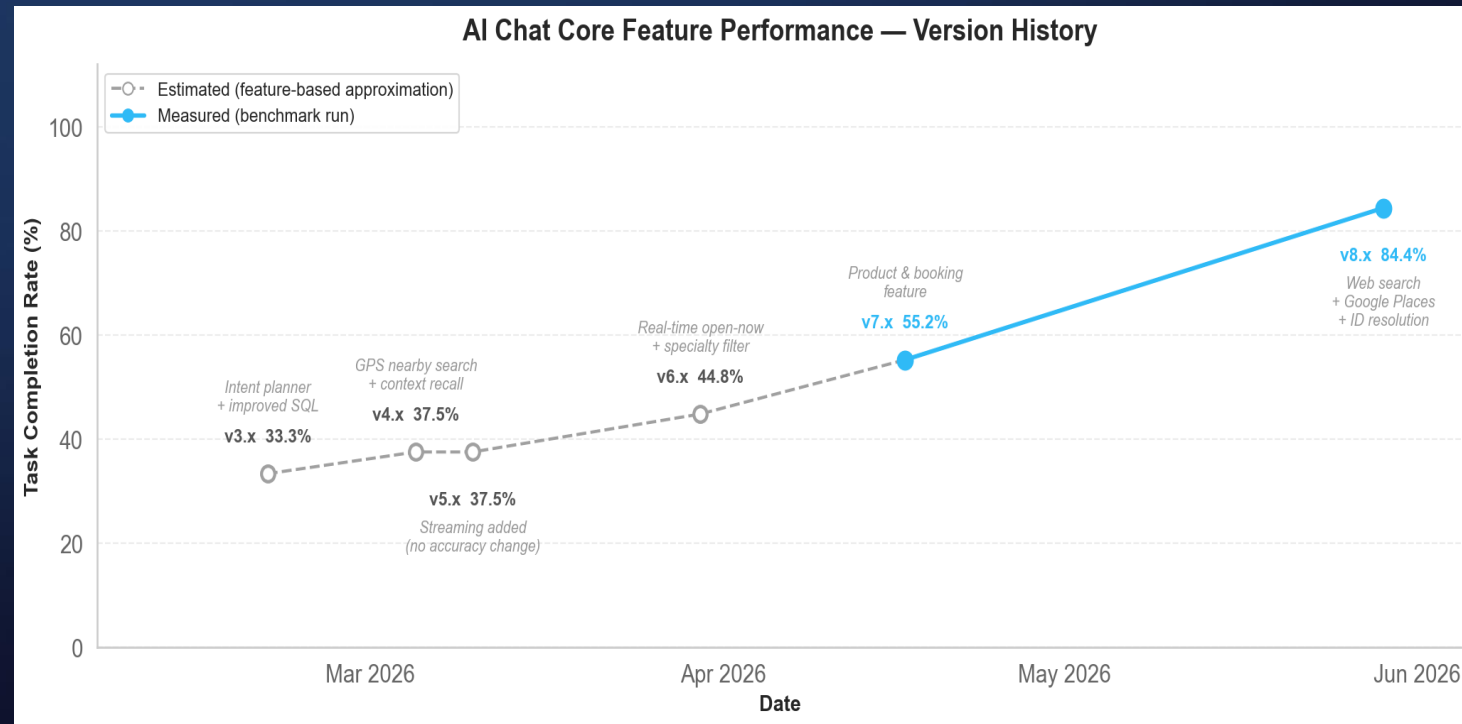
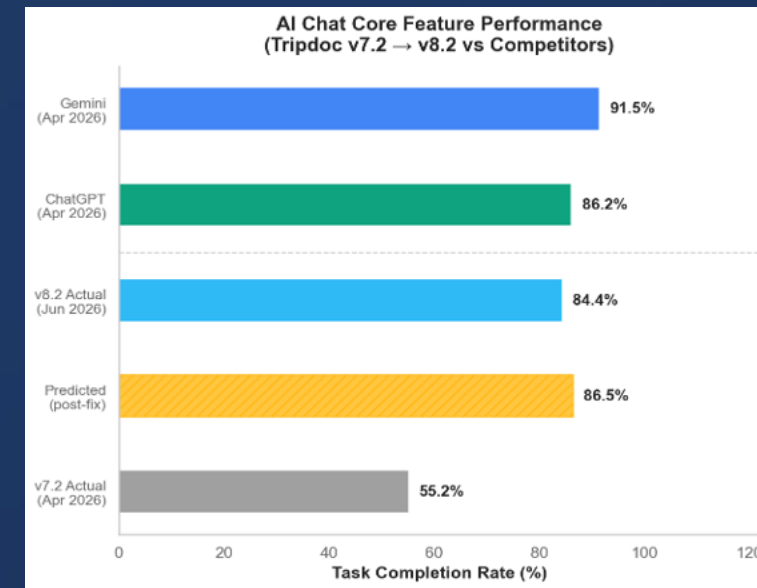
- Built and shipped from basic open source framework (OpenCHA) across several releases (v3 → v8), each adding production capability, along with fixes within 5 months.
- Development Tools: Python, Gemini API (runtime), SQL (DB), GCP (server), Docker (API deployment); assisted by AI coding tools (Claude, Codex)



PROJECT RESULTS:

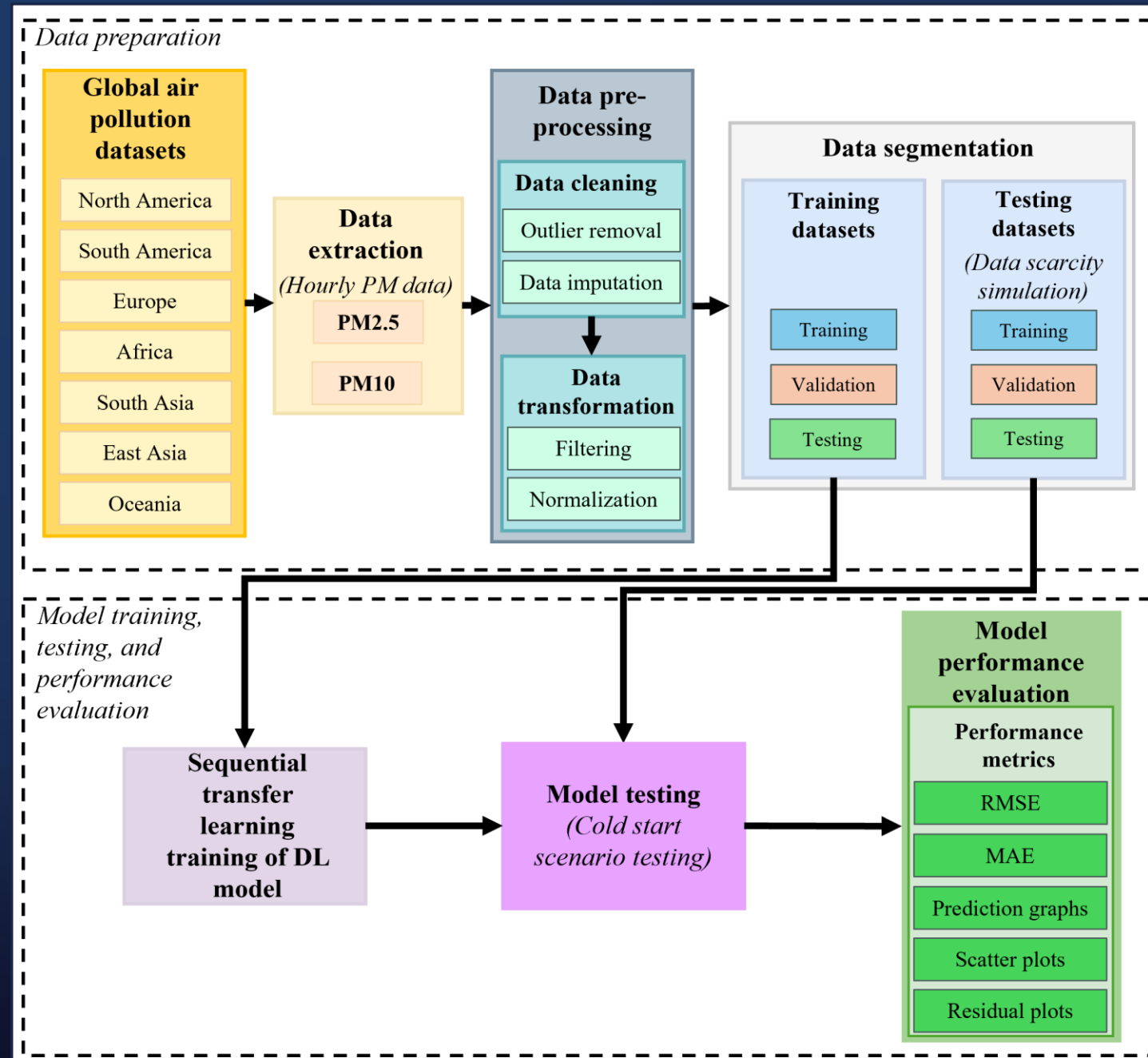
TRIPDOC AI CHAT ASSISTANT

- Designed and executed a 25-scenario multilingual benchmark evaluating the assistant against Gemini and ChatGPT; diagnosed structural failure modes and drove fixes.
- Raised task-completion rate from 55% → 84% across the benchmark.**
- Produced evaluation and roadmap reports converting findings into a prioritized, scheduled engineering plan for production readiness.



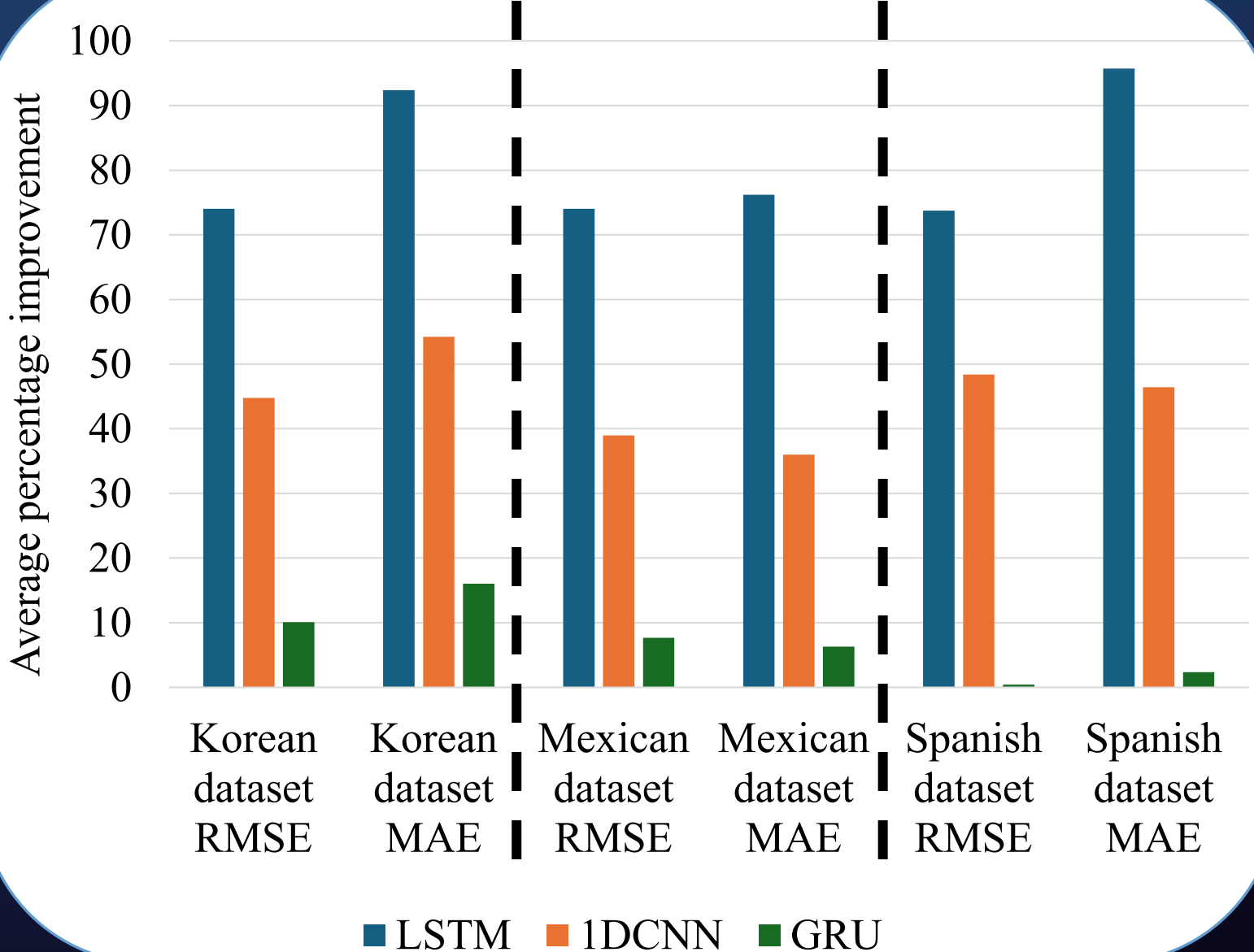
MASTER'S RESEARCH PROJECT: Global air pollutant forecasting using Sequential Transfer Learning

- The aim of this project was to predict air pollution levels anywhere in the globe via AI models.
- This would be useful in areas with limited air pollution measurement infrastructure and can also aid in public health management.
- Transfer learning was applied using air pollution data from data-rich regions to predict air pollution in areas with limited or no data.



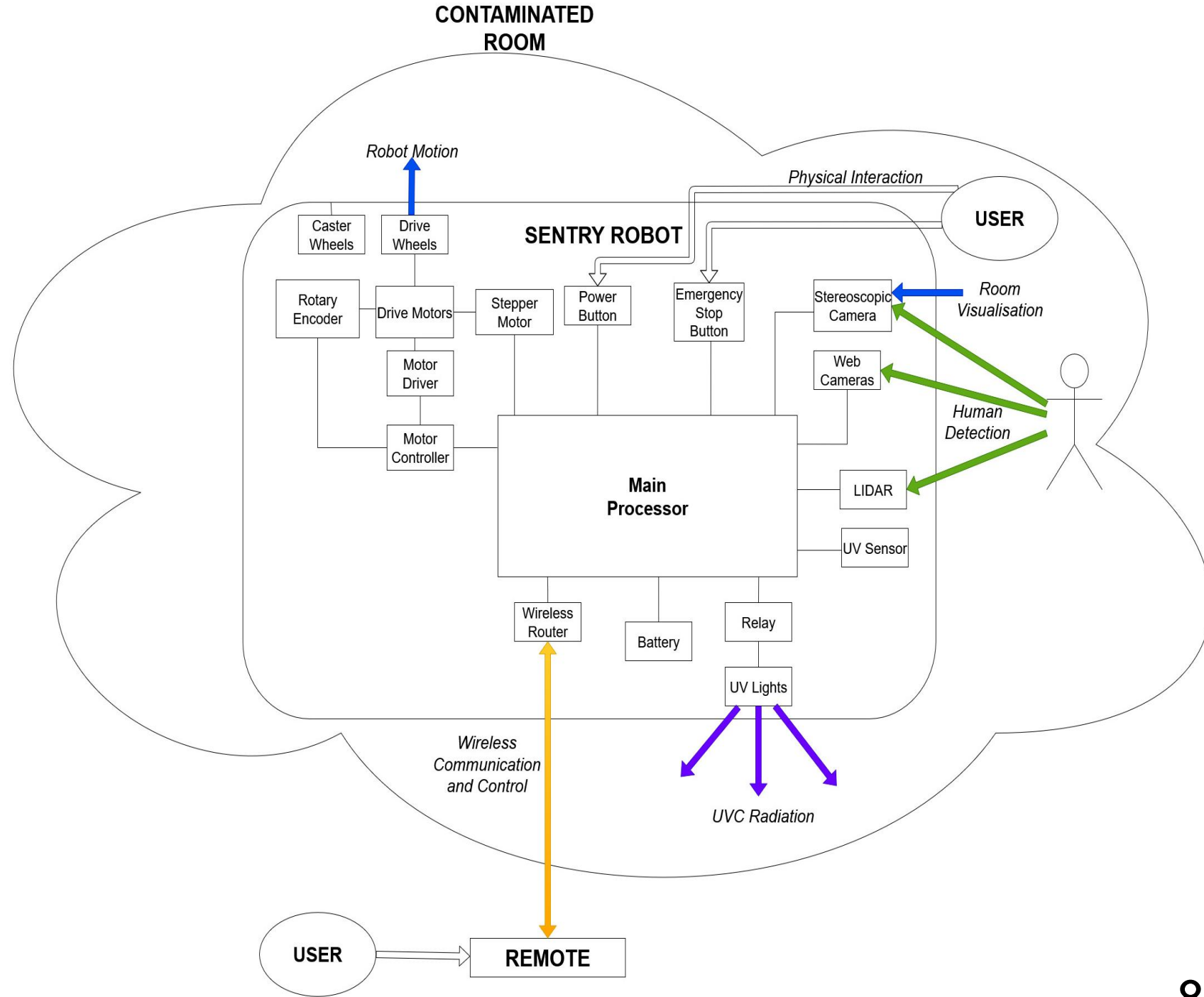
PROJECT RESULTS

- The developed transfer learning models were compared to an untrained baseline model across various data scarcity levels (0.5 weeks – 3 months of data).
- Three machine learning models (LSTM, 1DCNN, and GRU) were tested in three regions: Korea, Mexico, and Spain, to show the global performance of the applied method.
- On average, performance improved by 75% to 99% with RMSE ranging from 0.187 down to 0.0032 across the various data scarcity levels.
- Development Tools: Python, Pytorch



R&D PROJECT: DISINFECTION ROBOT

- During the COVID-19 pandemic, the government of Trinidad and Tobago sought a solution for cleaning hospital rooms.
- An autonomous disinfection robot using UV light was proposed as a solution.
- The robot would also be controlled wirelessly via a remote/ dedicated app.





ROBOT PROTOTYPE

- During the project I was responsible for creating Architecture diagrams and supported the design and implementation of the robot drivetrain.
- An Arduino was used for the motor controller and PD control was implemented to simultaneously control each motor.
- Navigation commands were transmitted from the main system controller via UART.
- Development Tool: Python

OTHER PROJECTS

- Construction Site Hole Detection Using AI
 - Development Tools: Python, Pytorch, YOLOv8
- Air Disinfection Device Control System
 - Development Tools: Python, Arduino
- Robotics Kit Upgrade
 - Development Tools: C#/ C++/ C, PIC

